

Korean Text Rendering in Text-to-Image Models: A Reproducible Character-Error-Rate Benchmark

Han Kim

IOV Labs (아이 오브 연구소) · hankim.masion@gmail.com · ORCID 0009-0000-5998-1358

Open benchmark · snapshot 2026-05-29 · compiled 2026-05-30

Code MIT · text and results CC BY 4.0 · one-command reproducible · archived on Zenodo

Abstract. Benchmarks for text inside generated images are overwhelmingly English, which conceals the writing systems where models actually fail. We measure one of them directly. Nine text-capable text-to-image models are each asked to draw fourteen Korean (Hangul) phrases on an identical plain poster; the rendered text is transcribed by a vision-language model (GPT-4o) and scored by **character error rate** (CER). The result is a sharp ranking and one blunt failure. Three models — recraft-v4-pro, seedream-5, and nano-banana-pro — render every prompt perfectly (CER 0.000, 14/14 exact), and a clear quality gradient follows. At the bottom, **imagen-4 cannot write Hangul at all**: it produces plausible-looking Korean-shaped gibberish on every prompt (0/14, mean CER 1.33), turning 커피 한잔 (“a cup of coffee”) into 소동석 고려아는 아래해안. The central finding is that **strong English text rendering does not transfer to Korean**, and is invisible to an English-only benchmark. The harness is open, runs with a single command, resumes from saved results, and is trivially extensible to new prompts and models.

Keywords: text-to-image generation · visual text rendering · Hangul · Korean · OCR · character error rate · evaluation · benchmark · reproducibility

Contributions.

1. A reproducible, single-command benchmark of **Korean** text rendering across nine current text-to-image models, with a character-error-rate metric and exact-match rate.
2. Evidence that English text-rendering quality **does not transfer** to Hangul: a top English renderer (imagen-4) scores 0/14 on Korean.
3. A qualitative error taxonomy over Hangul-specific failure modes — complex jamo clusters, tense consonants, digit drift, and long or uncommon strings.
4. An open harness (prompts, models, OCR scoring) that resumes from saved results and is easy to extend, archived with a DOI.

Contents

1	Introduction	3
1.1	Why Korean is a good stress test	3
1.2	Contributions and roadmap	3
2	Background and related work	3
2.1	Text-to-image generation	3
2.2	Visual text rendering	4
2.3	Evaluation: OCR, edit distance, and model-as-judge	4
3	Benchmark design	4
3.1	Task	4
3.2	Prompts	4
3.3	Models	4
3.4	Scoring	4
4	Results	5
4.1	Leaderboard	5

4.2	Three perfect renderers	6
4.3	The competent gradient	6
4.4	The blunt failure	6
5	Error analysis	7
6	Discussion	8
6.1	English skill does not transfer to Korean	8
6.2	OCR is itself a model	8
6.3	Implications	8
7	Limitations and threats to validity	8
8	Reproducibility	9
9	Conclusion	9
	References	9
A	Prompt set	11
B	Per-prompt character error rate	11
C	Metric definition	11

1 Introduction

Modern text-to-image models can place legible words inside an image — a storefront sign, a poster headline, a product label. This capability has improved quickly for English [1], [2], and recent systems render short English strings nearly flawlessly. But “the model can render text” is a claim almost always evaluated in English, and that choice quietly hides the writing systems where the same models break. A model that writes perfect English signage may be unable to draw a single correct Korean word, and an English-only benchmark will never reveal it.

This paper measures Korean (Hangul) text rendering directly, reproducibly, and across a current field of nine models. The task is deliberately narrow and unambiguous: draw a given Korean phrase as the only text on a plain white poster, in black sans-serif lettering. Because the target string is known, rendering quality can be scored objectively — we transcribe what each model actually drew with a vision-language model and compute the character error rate against the intended text. The narrowness is the point: by stripping away style, composition, and semantics, the benchmark isolates the one capability of interest.

The headline result is stark. Three of the nine models render all fourteen prompts perfectly. A clear gradient of partial competence follows. And one model widely regarded as a strong English renderer, imagen-4, fails **completely** on Korean — not with occasional typos, but by emitting plausible-looking Hangul-shaped nonsense on every prompt. The finding generalizes beyond a single model: **visual text-rendering skill is script-specific and does not transfer from English to Korean**, and the only way to know whether a model can write a given script is to measure it in that script.

1.1 Why Korean is a good stress test

Hangul is featural and compositional: each syllable block is assembled from an initial consonant (초성), a medial vowel (중성), and an optional final consonant (받침), drawn from a set of jamo that includes doubled “tense” consonants (ㄲ, ㄸ, ㅃ, ㅆ, ㅉ) and complex final clusters (ㄺ, ㄻ, ㄼ, ㄽ, ...). A renderer must place several sub-glyphs in the correct spatial arrangement **within** a single square block, then sequence the blocks. This is a more structured target than a Latin letter string, and it exposes failure modes — dropped or swapped jamo, collapsed clusters, mis-stacked blocks — that have no English analogue. Korean is therefore an informative probe of whether a model has learned glyph composition or merely memorized common English word-images.

1.2 Contributions and roadmap

Our contributions are listed above. Section 2 reviews text-to-image generation, visual text rendering, and evaluation. Section 3 specifies the benchmark — prompts, models, generation protocol, OCR transcription, and the CER metric. Section 4 reports the leaderboard. Section 5 gives the Hangul-specific error taxonomy with examples. Section 6 discusses the non-transfer finding and the limits of OCR-as-judge. Section 7 states threats to validity, Section 8 covers reproducibility, and Section 9 concludes. Appendices give the prompt set and the metric definition.

2 Background and related work

2.1 Text-to-image generation

Contemporary text-to-image systems are largely diffusion models conditioned on a text encoder [3], [4], [5]. Early systems were notoriously poor at rendering legible text inside images, a limitation often attributed to the text encoder’s tokenization and to the scarcity of glyph-level supervision. The models evaluated here are recent commercial and open systems served through a common API; we treat them as black boxes and measure only their output.

2.2 Visual text rendering

A focused line of work has improved in-image text specifically. Liu et al. [1] show that **character-aware** text encoders — which see characters rather than only sub-word tokens — substantially improve visual text rendering, directly implicating tokenization as a cause of failure. Glyph- and layout-conditioned methods such as GlyphControl [6] and TextDiffuser [2] add explicit glyph or position control to place text more reliably. This literature is overwhelmingly evaluated on English (and occasionally Chinese); Korean is rarely measured, despite Hangeul’s compositional structure making it an especially clean test of glyph-level competence. Our benchmark fills that specific gap with a direct, reproducible measurement.

2.3 Evaluation: OCR, edit distance, and model-as-judge

Rendering quality is naturally scored by reading the produced text and comparing it to the target. The comparison is an edit-distance problem: the Levenshtein distance [7] counts the minimum single-character insertions, deletions, and substitutions to transform one string into another, and normalizing it by target length yields the **character error rate** (CER), a standard measure in OCR and speech recognition. The reading step itself uses a model — here a vision-language model transcribes the image — which connects to the broader “model-as-judge” paradigm [8] and inherits its central caveat: the judge can be wrong, so the metric is a calibrated proxy, not ground truth (Section 6.2). Reference-free image-text metrics such as CLIPScore [9] measure semantic alignment, not character-exact rendering, and are therefore unsuitable for this task; exact transcription plus CER is the appropriate instrument.

3 Benchmark design

3.1 Task

For each (model, prompt) pair the model receives an identical instruction: produce a white poster whose only text is the target Hangeul string, set in black sans-serif lettering. Fixing the surface — white background, single black string, no decoration — isolates **text rendering** from style, layout, and semantics, so that any error is attributable to the model’s ability to draw the requested glyphs rather than to compositional choices.

3.2 Prompts

The fourteen prompts span easy-to-hard Hangeul, chosen to exercise the script’s structure rather than its vocabulary (Appendix A). They range from a short greeting (안녕하세요) and word-spacing (커피 한 잔) through deliberately hard cases: rare final clusters (값을 매기다 with ㅂ, 맑음 with ㄹ), tense consonants (떡볶이), a complex cluster plus tense consonant (닭갈비 맛집, ㄹ), a longer brand string (주식회사 아이오브), a full sentence (안녕하세요 반갑습니다), place names (서울특별시 강남구), and digit-Hangeul mixes (9월 14일 토요일, 커피 2잔 주세요). The set is small by design — the goal is a sharp, reproducible probe, not exhaustive coverage — and is trivially extensible via the repository’s prompts.json.

3.3 Models

Nine text-capable models served through the fal.ai API were evaluated in the 2026-05-29 snapshot: recraft-v4-pro, seedream-5, nano-banana-pro, gpt-image-2, recraft-v4, gpt-image-1.5, flux-2-flash, ideogram-v3, and imagen-4. One image was generated per (model, prompt) cell, for 126 generations in total. Model endpoints and versions move over time; the numbers are therefore explicitly dated.

3.4 Scoring

Let g be the string the model actually rendered and t the target. A vision-language model (GPT-4o) transcribes the largest visible text in the image to obtain g . The character error rate is the length-normalized Levenshtein distance with whitespace removed:

$$\text{CER}(g, t) = \frac{\text{lev}(g', t')}{|t'|}, \quad (1)$$

where x' denotes x with whitespace stripped and lev is the Levenshtein edit distance [7]. $\text{CER} = 0$ is a perfect render; values near or above 1 indicate text bearing little or no relation to the target. For each model we report the mean CER over the fourteen prompts and the **exact-match rate**, the fraction of prompts rendered with $\text{CER} = 0$.



Figure 1: The same prompts drawn by the nine models (one frame of the cycling comparison); each label shows the model and its CER on that prompt.

4 Results

4.1 Leaderboard

Table 1 gives the full ranking. The field separates into three regimes: a perfect tier, a competent gradient, and a single complete failure.

Rank	Model	Mean CER	Exact
1	recraft-v4-pro	0.000	14 / 14 (100%)
2	seedream-5	0.000	14 / 14 (100%)
3	nano-banana-pro	0.000	14 / 14 (100%)
4	gpt-image-2	0.038	12 / 13 (92%)
5	recraft-v4	0.071	13 / 14 (93%)
6	gpt-image-1.5	0.083	12 / 14 (86%)
7	flux-2-flash	0.145	9 / 14 (64%)
8	ideogram-v3	0.302	5 / 14 (36%)
9	imagen-4	1.332	0 / 14 (0%)

Table 1: Korean text-rendering leaderboard, nine models × fourteen Hangul prompts (2026-05-29 snapshot via fal.ai). Lower mean CER and higher exact-match rate are better.

```

> node summary.mjs

Korean Text Rendering in Image Models - 9 models x 14 Hangul prompts

where models slipped (target -> what was actually drawn):
imagen-4 커피 한 잔 -> 소동석 고려아는 아라해안 CER 2.75 X
imagen-4 맑음 -> 웅반재다 CER 2.00 X
imagen-4 닭갈비 맛집 -> 정장주니 암방에아련 CER 1.80 X
imagen-4 값을 매기다 -> 조로여능 화상내라 CER 1.60 X
imagen-4 책갈피 -> 화비장컨 CER 1.33 X
imagen-4 사랑해 -> 찬방역보 CER 1.33 X
imagen-4 떡볶이 -> 문련차병 CER 1.33 X
ideogram-v3 주식회사 아이오브 -> 주식회사 아이오브 CER 0.00 OK
recraft-v4 닭갈비 맛집 -> 닭갈비 맛집 CER 0.00 OK

Leaderboard (lower CER is better)
1. recraft-v4-pro CER 0.000 ##### 14/14 exact
2. seedream-5 CER 0.000 ##### 14/14 exact
3. nano-banana-pro CER 0.000 ##### 14/14 exact
4. gpt-image-2 CER 0.038 ##### 12/13 exact
5. recraft-v4 CER 0.071 ##### 13/14 exact
6. gpt-image-1.5 CER 0.083 ##### 12/14 exact
7. flux-2-flash CER 0.145 ##### 9/14 exact
8. ideogram-v3 CER 0.302 ##### 5/14 exact
9. imagen-4 CER 1.332 ..... 0/14 exact

OCR: GPT-4o - CER ignores whitespace - reproduce: node run.mjs
>

```

Figure 2: The harness prints the full leaderboard and a per-model exact-match bar chart in the terminal, reproducible with one command.

4.2 Three perfect renderers

recraft-v4-pro, seedream-5, and nano-banana-pro rendered all fourteen prompts with zero character error, including the hard cluster and tense-consonant cases. Their existence is the most important positive result: **correct Hangul rendering is achievable today**, so the failures below are not an inherent limitation of the medium but a property of specific models.

4.3 The competent gradient

Between the perfect tier and the failure lies a smooth gradient. gpt-image-2 (0.038), recraft-v4 (0.071), and gpt-image-1.5 (0.083) miss only on the hardest one or two strings. flux-2-flash (0.145, 9/14) and ideogram-v3 (0.302, 5/14) degrade on complex jamo and longer or less common strings, in the systematic ways catalogued in Section 5. The gradient is informative: it shows that Hangul competence is partial and string-dependent, not a binary the model either has or lacks — except at the extremes.

4.4 The blunt failure

imagen-4 is the outlier and the paper’s central cautionary example. It scored 0 of 14 with a mean CER of 1.33 — above one, meaning its output is on average **further** from the target than an empty string would be, because it confidently emits wrong characters rather than omitting them. It does not produce Korean with occasional errors; it produces **Hangul-shaped nonsense** on every prompt: 커피 한 잔 → 소동석 고려아는 아라해안, and 맑음 → 웅반재다 (Figure Figure 3). The glyphs are individually well-formed Korean syllable blocks, which is what makes the failure striking — the model has learned what Hangul **looks like** without learning to render **specific** requested words.



Figure 3: imagen-4 on Korean prompts. Each panel’s label is the intended text; the image is what the model actually drew — well-formed but unrelated Hangul.

5 Error analysis

Beyond the aggregate scores, the **kinds** of errors are diagnostic of where glyph composition breaks. We group the observed failures (excluding imagen-4, whose output is unrelated to the target) into four Hangul-specific modes.

Complex final clusters (겹 받침). Syllables ending in a two-consonant cluster are a frequent failure point. The cluster ㅃ in 맑음 collapses to a single consonant — flux-2-flash renders 맑음 — losing one jamo from the block. These clusters require stacking two final consonants in a position that is itself already crowded, and weaker renderers drop or merge them.

Tense (doubled) consonants. Doubled consonants are mis-rendered as their single counterparts: 떡볶이 → 덕볶이 (flux-2-flash), where the tense ㅌ is read back as a plain ㄷ. The doubling is a fine visual distinction that lower-fidelity renderers smear.

Digit–Hangul mixing. Strings that mix Arabic numerals with Hangul are a soft spot for the weaker renderers: on the date prompt 9월 14일 토요일, flux-2-flash drifts (CER 0.375) while the perfect tier and even ideogram-v3 handle it. Switching scripts mid-string appears to destabilize the character count.

Long or uncommon strings. Less frequent or longer strings degrade most: the brand string 주식회사 아이오브 collapses in flux-2-flash (CER 0.625), dropping several syllables. Rarer word-images give the model less to memorize and more to compose, exposing weak composition.

The hardest single prompt, 닭갈비 맛집 — combining a complex final cluster (ㄹ in 닭) with a tense consonant — is shown across all nine models in Figure Figure 4. The spread on this one prompt mirrors the leaderboard: the perfect tier draws it cleanly, the gradient stumbles on the cluster, and imagen-4 produces unrelated glyphs.



Figure 4: The hardest prompt, 닭갈비 맛집, rendered by all nine models (label = model + CER). The complex cluster ㄹ and the tense consonant separate the field.

6 Discussion

6.1 English skill does not transfer to Korean

The single most important takeaway is a non-transfer result. imagen-4 is, by reputation and on English benchmarks, a strong text renderer; on Korean it is the worst model tested, unable to draw a single correct word. Visual text rendering is therefore **script-specific**: competence in one writing system says little about another, especially across a script boundary as large as Latin-to-Hangul. The most plausible mechanism is consistent with the character-aware-encoder finding of Liu et al. [1] — if a model’s text conditioning is tokenized in a way that fragments or under-represents Hangul jamo, the model can learn the **appearance** of Korean (well-formed syllable blocks) without learning to compose **specific** targets, exactly the imagen-4 signature. Whatever the cause, the practical consequence is unambiguous: a model’s English text score must not be assumed to hold for Korean, and the only reliable knowledge is a measurement in the target script.

6.2 OCR is itself a model

Our metric reads each image with a vision-language model, which makes the judge fallible — a manifestation of the model-as-judge caveat [8]. A misread inflates CER for a render that was actually correct, or (less often) the reverse. Two design choices bound the risk. First, the task is adversarial to false positives: the targets are short, the surface is clean, and a perfect render is unambiguous, so CER 0 is rarely awarded in error. Second, the failure that drives the headline — imagen-4 at mean CER 1.33 — is far outside any plausible OCR noise band; no transcription error explains a model that never produces the requested word. The gradient in the middle of the table is where OCR noise matters most, and those fine distinctions should be read as approximate.

6.3 Implications

For practitioners, the result is directly actionable: if an application renders Korean, the model must be chosen on a Korean measurement, and three models are shown to be safe choices today. More broadly, the benchmark is a concrete instance of a cheap automatic checker (OCR + CER) used as a **quality gate** and a **routing signal** — the same pattern the lab has argued for in generative-media verification: a fast, objective check that can both reject bad outputs and route a request to a model known to handle its script.

7 Limitations and threats to validity

We are explicit about what this benchmark does and does not establish.

The OCR judge is fallible. CER is a proxy computed from a model’s transcription, not ground truth; fine differences in the middle of the table are approximate (Section 6.2).

Small n . One image per (model, prompt) cell, with no sweep over seeds, aspect ratios, or prompt phrasings. This is a **snapshot**, not a verdict: a model near a boundary could move with a different seed. The fourteen prompts probe structure, not coverage of the language.

Endpoints move. Model versions and API paths change; the numbers are dated 2026-05-29 and should be re-run before being cited as current.

Surface specificity. The clean-poster surface isolates rendering but is not representative of in-the-wild prompts (busy scenes, stylized type); a model could render Hangul well in isolation yet poorly in a complex composition, or vice versa.

Single language. Korean is one informative script; the non-transfer claim is demonstrated from English to Korean and is not a general statement about every script pair, though it strongly motivates per-script measurement.

8 Reproducibility

The benchmark is open and runs with a single command given a fal.ai key (generation) and an OpenAI key (OCR):

```
export FAL_API_KEY=...      # image generation
export OPENAI_API_KEY=...   # OCR transcription (GPT-4o)
node run.mjs                # writes results.json + REPORT.md
node summary.mjs            # pretty-prints the saved leaderboard
```

A full run is roughly 126 generations (a few dollars of API). The harness **resumes from saved results**: re-running retries only failed cells, so a partial or interrupted run is cheap to complete. The prompt list (prompts.json) and the model list (in run.mjs) are the two extension points. The repository ships a CITATION.cff, is archived on Zenodo with a DOI, and licenses code under MIT and the writeup and results under CC BY 4.0.

9 Conclusion

We measured how well nine current text-to-image models draw Korean, with a narrow, objective, one-command benchmark: render a Hangul phrase on a plain poster, transcribe it, score it by character error rate. Three models are perfect, a clear gradient of partial competence follows, and one model widely held to be a strong English renderer — imagen-4 — cannot write Hangul at all, producing well-formed nonsense on every prompt. The lesson is general and easy to state: **visual text-rendering skill does not transfer across scripts, and you only learn whether a model can write Korean by measuring it in Korean.** The harness is open, dated, and reproducible, so the measurement can be repeated, extended to other scripts, and kept current as the models move.

Data and code availability. The harness, prompts, raw results (results.json, including image URLs), per-prompt report (REPORT.md), and this paper’s source are public in the project repository, archived on Zenodo with a DOI. Code is MIT-licensed; the writeup and results are CC BY 4.0.

Acknowledgements. Internal research of IOV Labs (아이오브연구소). Image generation via fal.ai endpoints; OCR via GPT-4o. Typeset with Typst; terminal figures recorded with vhs.

References

- [1] R. Liu *et al.*, “Character-Aware Models Improve Visual Text Rendering,” in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 16270–16297.
- [2] J. Chen, Y. Huang, T. Lv, L. Cui, Q. Chen, and F. Wei, “TextDiffuser: Diffusion Models as Text Painters,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

- [3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-Resolution Image Synthesis with Latent Diffusion Models,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10684–10695.
- [4] C. Saharia, W. Chan, S. Saxena, and others, “Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [5] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, “Hierarchical Text-Conditional Image Generation with CLIP Latents,” *arXiv preprint arXiv:2204.06125*, 2022.
- [6] Y. Yang *et al.*, “GlyphControl: Glyph Conditional Control for Visual Text Generation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [7] V. I. Levenshtein, “Binary Codes Capable of Correcting Deletions, Insertions, and Reversals,” *Soviet Physics Doklady*, vol. 10, no. 8, pp. 707–710, 1966.
- [8] L. Zheng, W.-L. Chiang, Y. Sheng, and others, “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [9] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, and Y. Choi, “CLIPScore: A Reference-free Evaluation Metric for Image Captioning,” in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021, pp. 7514–7528.

Appendix A — Prompt set

The complete fourteen-prompt set (from the repository’s `prompts.json`), ordered as in the harness, with the Hangul structure each one targets:

id	target	what it tests
greeting	안녕하세요	basic greeting (baseline)
spacing	커피 한 잔	word spacing
batchim	책갈피	final consonants (받침)
date	9월 14일 토요일	digits + Korean
phrase	행복을 드립니다	full phrase with 받침
doublecons	닭갈비 맛집	complex jamo cluster ㄹㅂ (hardest)
brand	주식회사 아이오브	longer brand string
short	사랑해	short, simple
value	값을 매기다	rare cluster ㅂㅅ
clear	맑음	rare cluster ㄹㅂ
tteok	떡볶이	tense consonants ㅌ / ㅍ
sentence	안녕하세요 반갑습니다	longer sentence
place	서울특별시 강남구	place names
order	커피 2잔 주세요	number + honorific

The set is intentionally small — a sharp probe of structure rather than broad lexical coverage — and is designed to be extended by users for their own targets.

Appendix B — Per-prompt character error rate

Full CER matrix (2026-05-29 snapshot). Columns are the nine models in leaderboard order; “—” marks a generation that failed to return an image. A perfect render is 0.000; values above 1 (imagen-4) indicate output further from the target than an empty string.

prompt	r4-pro	sd-5	nbp	gi-2	r4	gi-1.5	flux	ideo	im-4
greeting	0	0	0	0	0	0	0	0	1.00
spacing	0	0	0	0	0	0	0	0	2.75
batchim	0	0	0	0	0	0	0	0.67	1.33
date	0	0	0	0	0	0	0.38	0	0.50
phrase	0	0	0	0	0	0	0	0.29	1.00
doublecons	0	0	0	0	0	0	0	0.40	1.80
brand	0	0	0	0	0	0	0.63	0	1.00
short	0	0	0	0	0	0	0	0.33	1.33
value	0	0	0	—	0	0	0.20	0.20	1.60
clear	0	0	0	0.50	1.00	0.50	0.50	1.00	2.00
tteok	0	0	0	0	0	0.67	0.33	0.67	1.33
sentence	0	0	0	0	0	0	0	0.30	1.00
place	0	0	0	0	0	0	0	0.38	1.00
order	0	0	0	0	0	0	0	0	1.00

Columns: recraft-v4-pro, seedream-5, nano-banana-pro, gpt-image-2, recraft-v4, gpt-image-1.5, flux-2-flash, ideogram-v3, imagen-4. The single hardest prompt is **clear** (맑음, cluster ㄹㅂ), the only one to trip a member of the otherwise-strong middle tier; imagen-4 fails uniformly.

Appendix C — Metric definition

For target t and rendered (transcribed) string g , let t' , g' be the strings with all whitespace removed. The Levenshtein distance $\text{lev}(g', t')$ is the minimum number of single-character insertions, deletions, and substitutions transforming g' into t' [7]. The character error rate is

$$\text{CER} = \frac{\text{lev}(g', t')}{|t'|}, \quad (2)$$

which is 0 for a perfect render and can exceed 1 when the rendered string is both wrong and not shorter than the target (the imagen-4 regime, mean 1.33). The **exact-match rate** for a model is the fraction of its prompts with $CER = 0$. Mean CER averages Equation 1 over the fourteen prompts. Whitespace is ignored because poster line-breaking is a layout choice orthogonal to whether the correct characters were drawn.